AX88772E USB2.0 to Fast Ethernet Application Design Note

Revision 1.10
September 30th, 2026

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Revision History

Revision	Date	Description
V1.00	2026/08/14	Initial release
V1.10	2026/09/30	Modified some description in Section 5.

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1. Introduction

The AX88772E USB 2.0 to Fast Ethernet controller is a high performance and highly integrated ASIC which enables low cost, small form factor, and simple plug-and-play Fast Ethernet network connection capability for desktops, notebook PC's, Ultrabook's, docking stations, game consoles, digital-home appliances, and any embedded system using a standard USB port

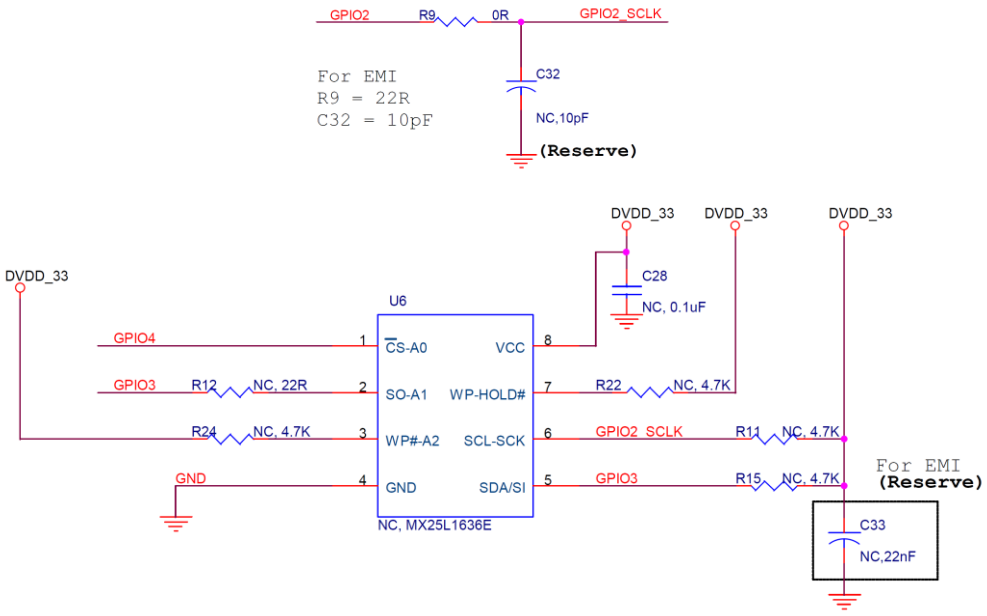
This Application Design Note provides important information about external component selection, schematic design and PCB design/layout for designing with ASIX Electronics' USB2.0-to-Fast Ethernet solution. ASIX Electronics highly recommends that user read through this Design Note before starting hardware design on schematic capture and PCB layout.

Note that this design note uses the AX88772E reference schematic for examples. Please refer to AX88772E reference schematics for exact reference circuits.

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2. External SPI Flash Selection Considerations

The following is the SPI Flash reference circuit of AX88772E USB 2.0 to Fast Ethernet controller. The SPI Flash is option for AX88772E.



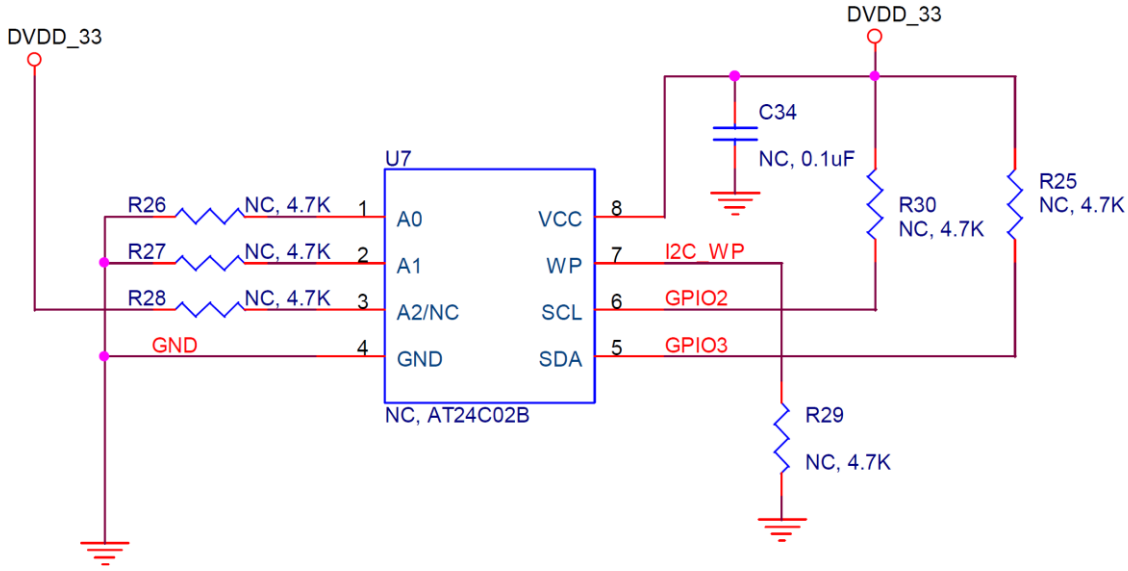
Note:

1. The AX88772E supports external SPI flash, the flash SCLK frequency must be supported 40MHz for READ instruction and size 1Mbit above.
2. The R9 and C32 must be closed to AX88772E's GPIO2 pin.
3. The SPI flash must be closed to AX88772E, the path distance of GPIO2 as short as possible

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3. I2C EEPROM Selection Considerations

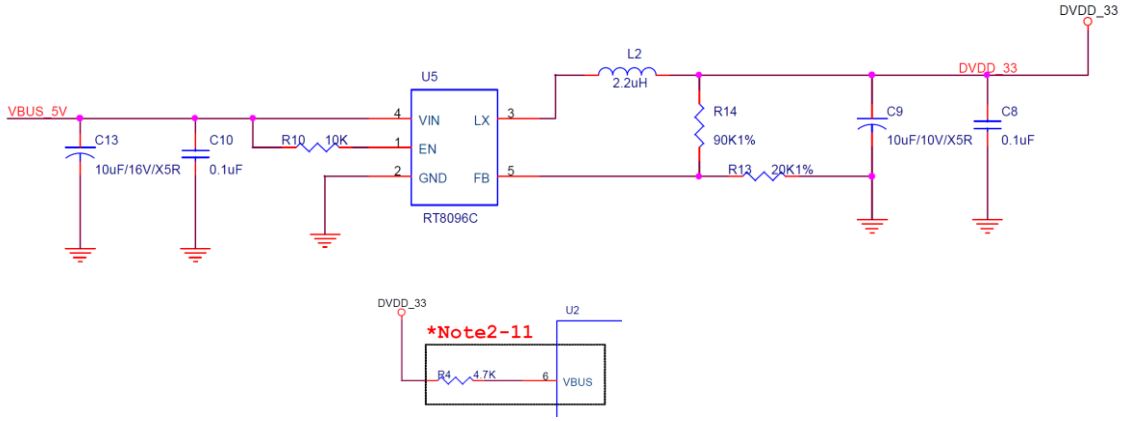
The following is the I2C EEPROM reference circuit of AX88772E USB2.0 to Fast Ethernet controller and the reference I2C EEPROM parts. The EERPOM is option for AX88772E.



Note: The EEPROM address must be 100.

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4. VBUS pin Design Considerations



VBUS	I3/S	6	VBUS signal of USB.

The AX88772E VBUS is a 3.3v pin input, it shall connect to the VBUS signal of USB connector for judging USB is attach or detach, AX88772E VBUS pin should be at low if AX88772E device is detached. e.g., If your application is connected to next layer of USB hub, the VBUS pin of AX88772E shall connect to VBUS pin of USB hub. If your application is on the same PCB (embedded system), the VBUS pin of AX88772E shall connect to controllable GPIO of MCU.

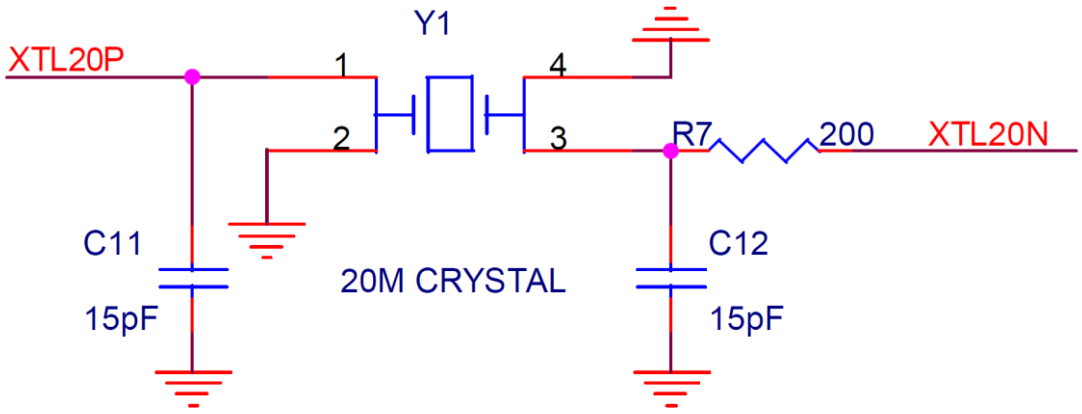
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5. Crystal Selection Considerations

The following is the specification of NSK NXK20.000AC20F-BK4 SMD 20Mhz crystal with CL 20pF and Drive Level 10uW, for AX88772E USB 2.0 to Fast Ethernet controller.

Parameter	Symbol	Typical Value
Nominal Frequency	Fo	20.000000MHz
Oscillation Mode		Fundamental
Frequency Tolerance (@25℃)		±30ppm
Frequency Stability Over Operating Temperature Range		±30ppm
Equivalent Series Resistance	ESR	100 Ohm max.
Load Capacitance	CL	20pF
Drive Level		10uW
Operation Temperature Range		-20℃ ~ +85℃
Aging		±3ppm/year

The following is an example of the 20MHz crystal clock circuit.



Note:

1. Please make sure the XTL20N 20MHz clock output signals are within 20MHz +- 30ppm. If the XTL20N 20Mhz clock output signals are out of the specification, please fine-tune the load capacitors (C11, C12) to meet the specification.
2. AX88772E does not support external oscillator clock source.

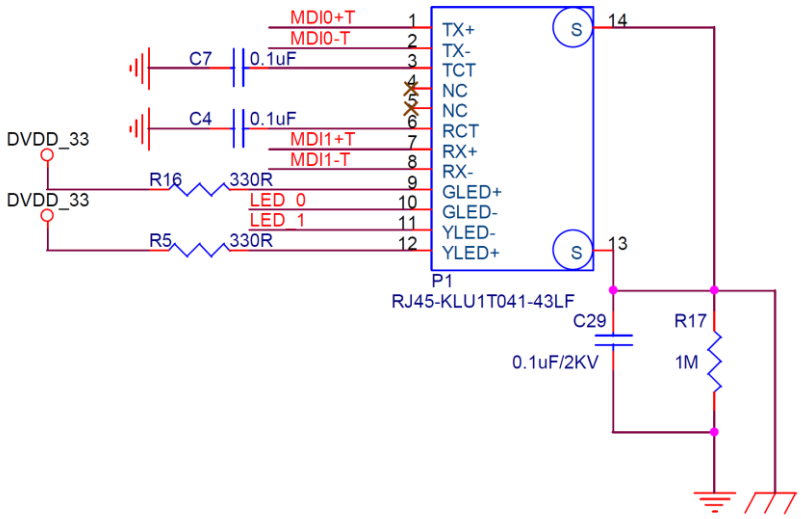
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6. Ethernet Magnetic Selection Considerations

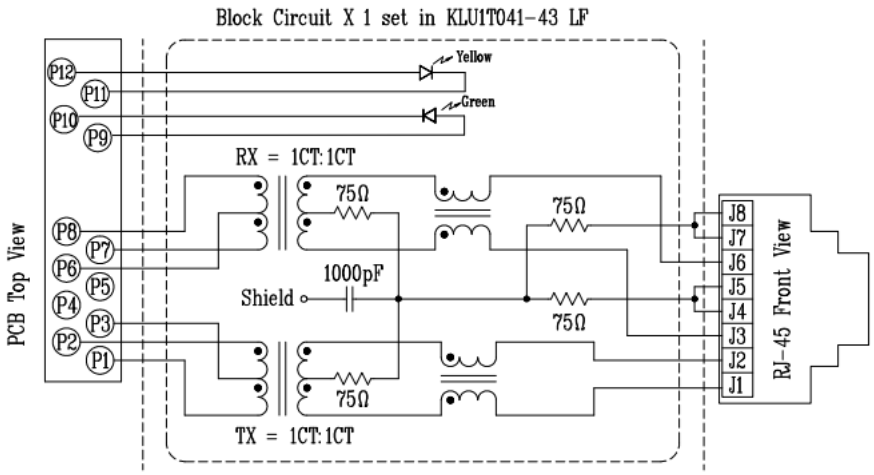
Please refer to AX88772E reference schematics for exact reference circuits.

6-1 Single RJ-45 Connector (with integrated magnetic)

The following is an example reference circuit of single RJ-45 connector (with integrated 10/100 Base-TX magnetic and Turns Ratio 1CT:1CT) on AX88772E demo board.



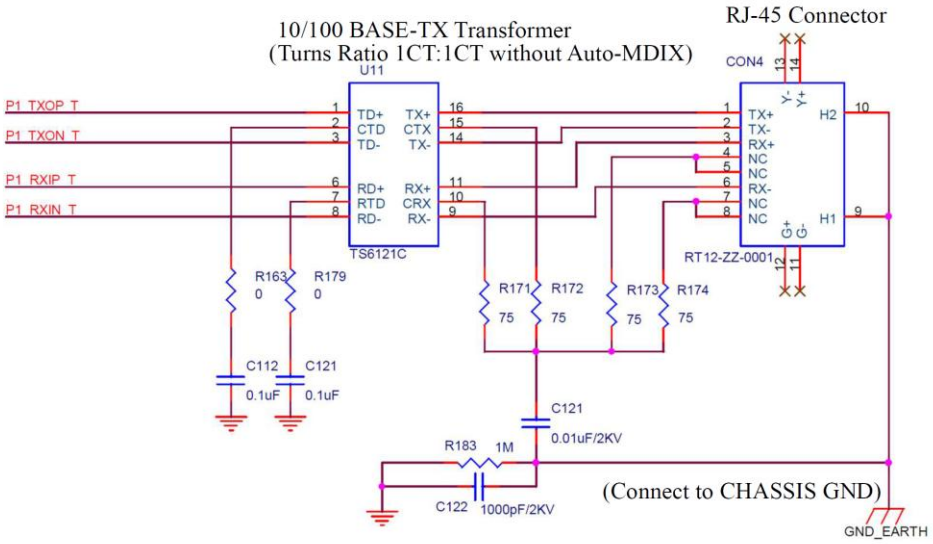
The following is an example (Bothhand KLU1T041) of single RJ-45 connector (with integrated 10/100 Base-TX magnetic).



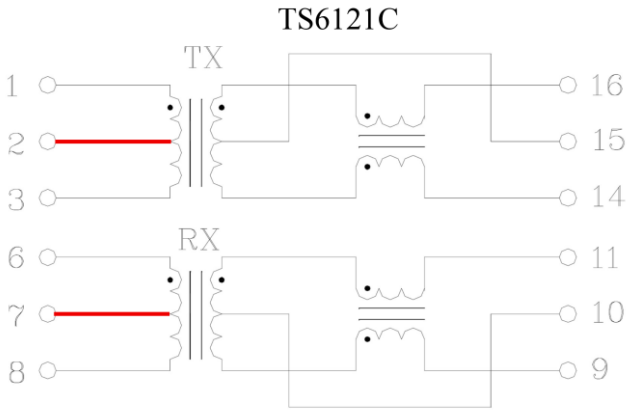
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6-2 Separate 10/100M Base-TX Magnetics and RJ-45 Connector

The following is an example reference circuit of separate 10/100M Base-TX magnetics and RJ-45 connector (Turns Ratio 1CT:1CT).



The following is an example (Bothhand TS6121C) of 10/100 Base-TX Ethernet transformer.



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7. 4-Layer PCB Design

We strongly suggest customers to design ASIX Electronics' USB-to-LAN Ethernet controller on the printed circuit board (PCB) with at least 4 layers. The 4-layer PCB design can help reduce some potential EMI, thermal and signal integrity issues, etc.

The following is an example 4-Layer PCB design for an embedded system application that uses ASIX Electronics' USB-to-LAN Ethernet controller.

Layer 1	Component (Top)	Magnetic and major signals
Layer 2	Ground	Digital/analog ground planes
Layer 3	Power	Digital/analog power planes
Layer 4	Component (Bottom)	Magnetic and other signals

Figure 1. An Example of 4-Layer PCB Design

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8. Power and Ground Planes Considerations

1. Digital GND and Analog GND can be placed together in one-piece so as to enlarge the GND, and thus helping dissipation of noises.

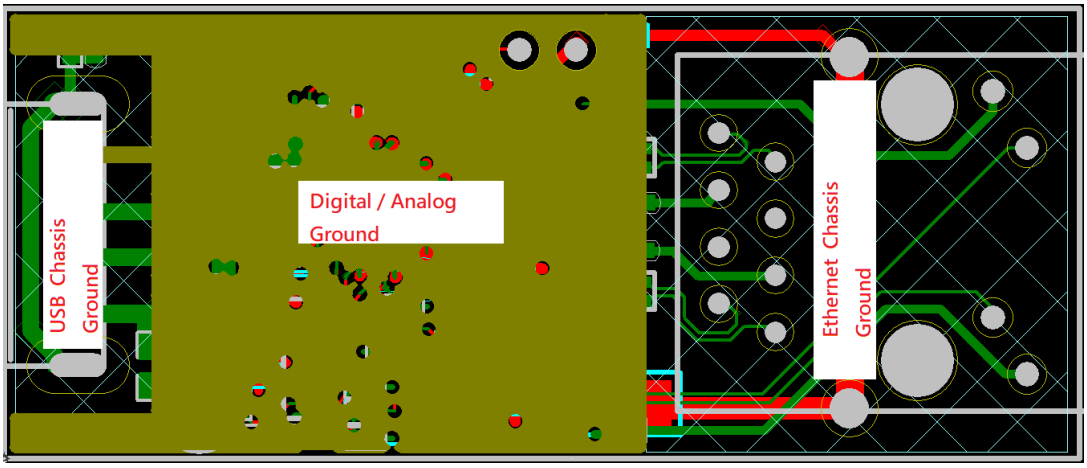
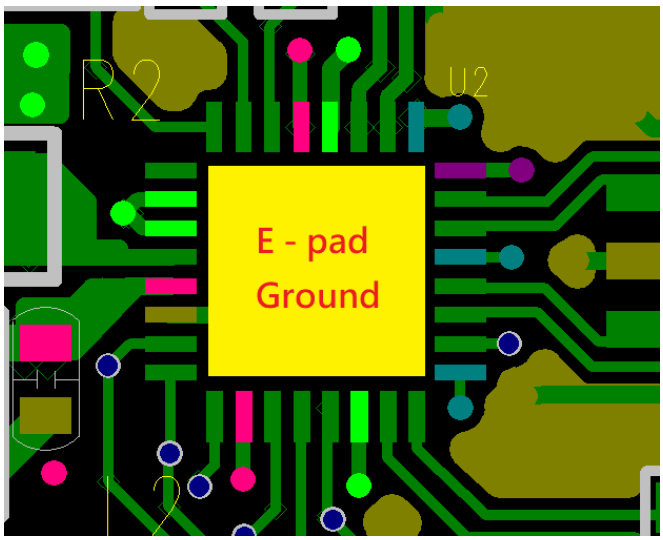


Figure 2. Typical Chassis/Digital Ground Planes

2.Connect AX88772E GNDK E-pad to digital ground plane to reduce AX88772E operating temperature.



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3. All the digital and analog power planes for different voltage supplies should be handled separately with different blocks to supply power to IC chip and peripheral components.

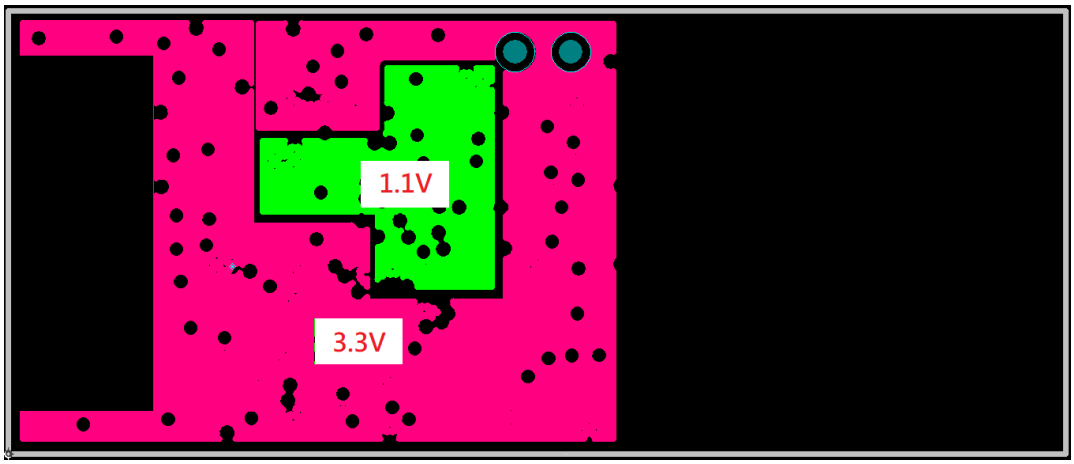
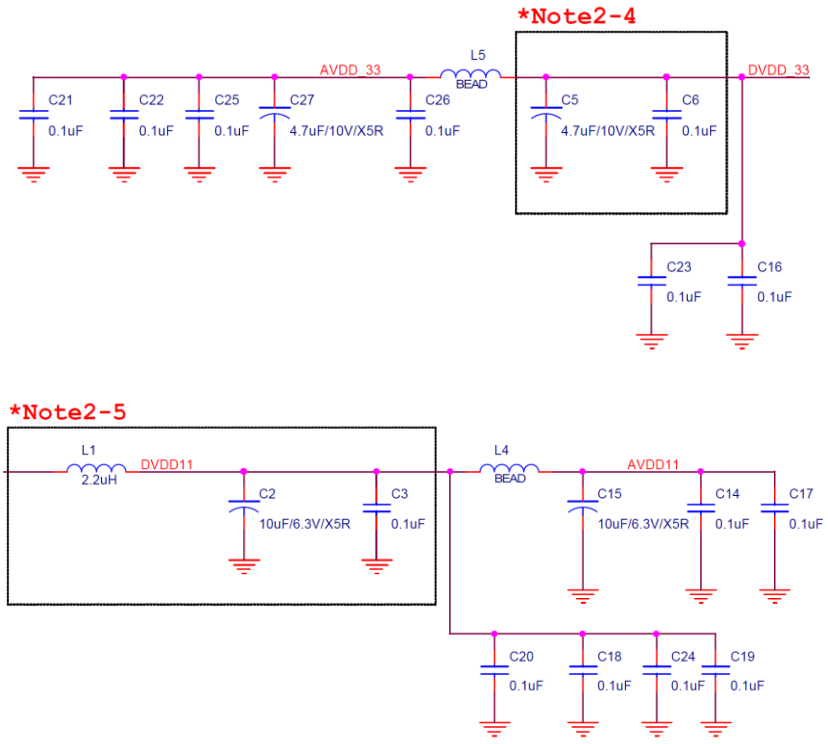


Figure 3. Typical Digital/Analog Power Planes

Note: The above figures are the Digital and Analog Power Planes diagram of an illustrative LAN board design. For exact layout pattern, ASIX Electronics provides AX88772E demo board and the layout PCB files and Gerber files for customer reference.

4. The digital and analog power planes should be isolated with a Ferrite Bead to isolate the noise source, and all power planes should be implemented with a large compensating capacitor to provide a stable power source.



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5. All power pins should be implemented with a decoupling capacitor, and the decoupling capacitor should be as close to the respective power pin of ASIX Ethernet controller as possible.

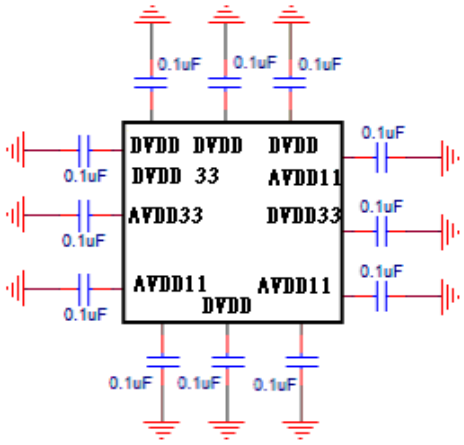
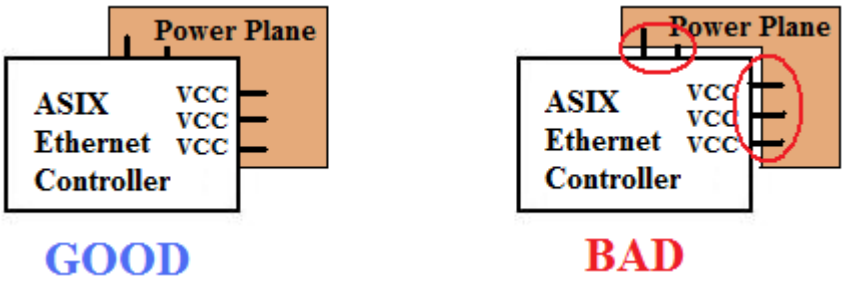


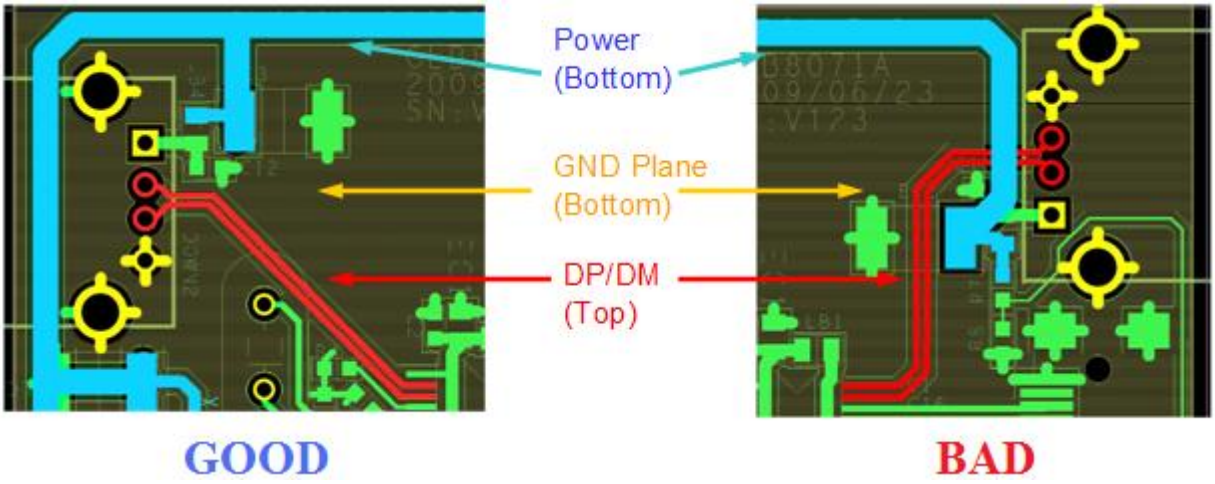
Figure 4. An Example of Power Pins and Decoupling Capacitors Circuits

6. Provide a power plane right underneath the ASIX Ethernet controller such that the VCC pins can be contacted to the power plane without going through thin traces.

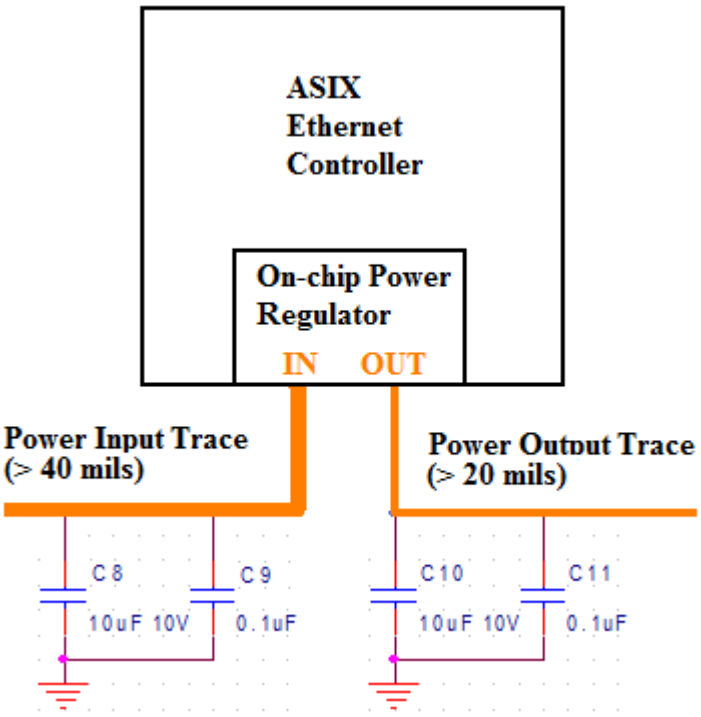


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7. Keep GND plane with no interruptions when routing power traces.



8. The on-chip regulator power input and output traces should be wider than other signal traces for lower power trace impedance. The power input trace is suggested to be wider than 40 mils and the power output trace is wider than 20 mils. The C8/C9 and C10/C11 compensating capacitors should be as close as the on-chip regulator input/output pins for good power efficiency.



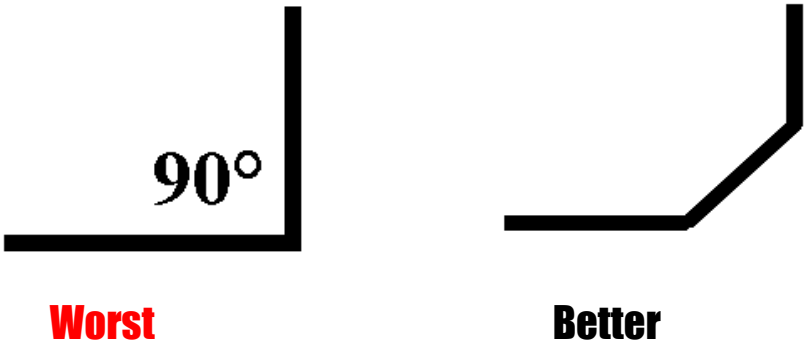
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9. USB Layout Guideline

This section describes some general USB signal layout guideline for the differential signals of USB 2.0 interface and the USB connector.

9-1 USB Layout Notes

1. The ideal USB differential impedance between D+ and D- traces should be 90 Ohm. The USB differential impedance between the D+/D- trace and GND should be 45 Ohm respectively. The designer can use the trace width of D+/D- traces to fine tune the USB differential impedance value (a wider trace has a smaller impedance value). If the PCB manufacturer provides TDR measurement, it helps control the PCB Impedance.
2. The differential signals, D+/D- should be routed with minimum trace length. Layout consideration should carefully avoid the nearby clock or noisy digital signals being routed too close to the D+/D- signals.
3. The USB differential signals between USB connector and DP/DM should be routed on the same component side on PCB, adjacent to the layer for ground plane. Avoid routing these differential signals through some vias or through ground plane to inadvertently affect the impedance control on these high-speed differential traces.
4. Any 90-degree turn in trace routing should be accomplished with two 135-degree turns as shown in example below.

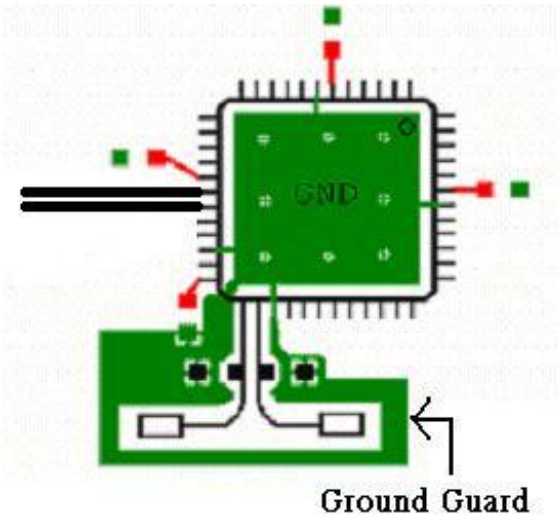


5. Avoid routing the D+ and D- near crystal, clock synthesizer, transformer, and other IC with switching noise from digital signals.
6. Avoid routing the D+ and D- near the edge of PCB or power planes.
7. The trace spacing between D+ and D- is 8 mils.
8. Keep the trace length of D+ and D- as equal as possible.The USB 2.0 differential pairs should not exceed 50 mils relative trace length difference
9. Wires for D+, D- shall be parallel and guarded by GND.
10. Forbid any Power Trace or Clock Line to pass underneath a Chip.
11. All Beads are to be placed as near to the USB connector as possible.
12. All bypass and electrolytic capacitors are to be placed as near the IC as possible.
13. All coupling capacitors of D+/D- shall be placed as near the IC as possible.

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14. Avoid punch-through holes at: USB differential signals paths (D+, D-). A punch-through hole may cause variation of signal line impedance, and distortion of signal in severe case. If it is inevitable to make such holes, the waveform will be inferior to that without the holes. The client shall make their own decision.

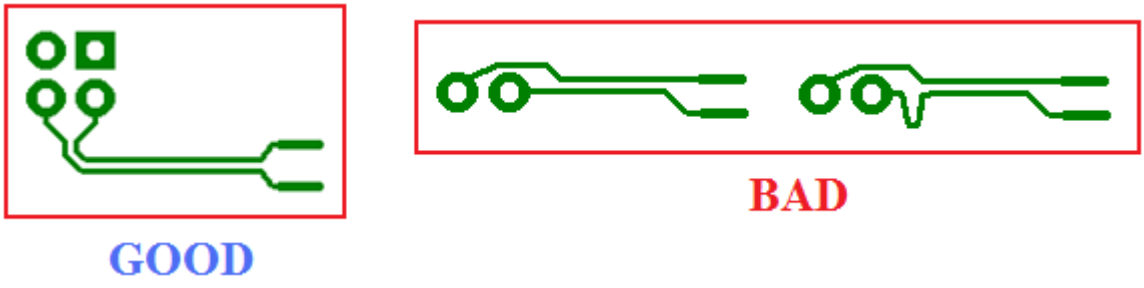
15. The X'tal (oscillator) traces shall be surrounded by GND to prevent interference to other signal wires. Better EMI effect can be attained by a one-piece GND underneath the IC furnished with PTH holes to enlarge the GND area.



16. The X'tal traces shall be symmetrical and parallel in order to get better oscillation wave form and better EMI protection.

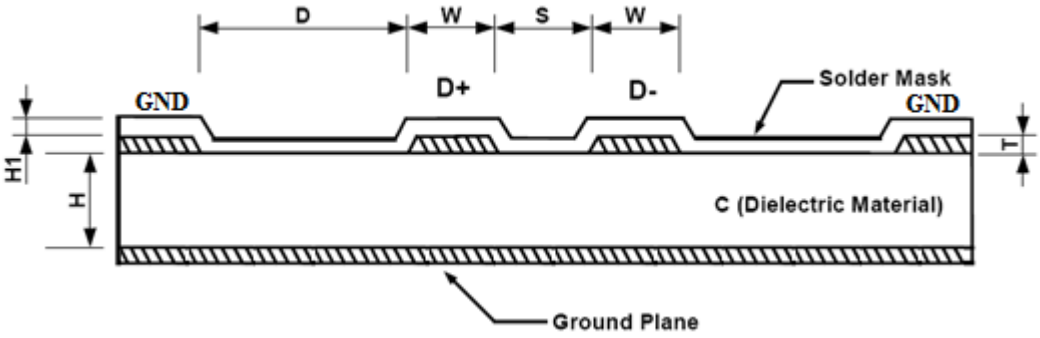


17. USB signal traces shall be placed parallel as below pictures.



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9-2 PCB (FR4 material) and impedance



1. W = 8 mils
2. S (D+ to D-) = 8 mils
3. T = Thickness of the trace = 1 ounce copper
4. D (Ground Separation) ≥ 20 mils
5. H (Dielectric thickness, distance of trace from the ground plane. Board thickness) ≈ 63 mils. The H value is dependent on the PCB material and might need to be fined tune based on your real PCB material.
6. The ideal USB differential impedance between D+ and D- traces should be 90 Ohm. The USB differential impedance between the D+/D- trace and GND should be 45 Ohm respectively. The designer can use the trace width of D+/D- traces to fine tune the USB differential impedance value (a wider trace has a smaller impedance value). If the PCB manufacturer provides TDR measurement, it helps control the PCB Impedance.

9-3 Trace route for USB2.0 PHY and Connector

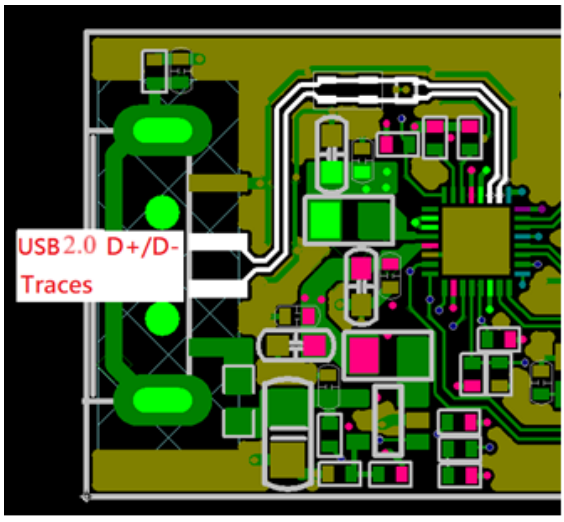


Figure 5. The Trace Route between the AX88772E and the USB connector

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10.Ethernet Magnetic Layout Considerations

This section describes some general Ethernet layout considerations for the differential signals of the Ethernet controller, the Ethernet magnetic and the RJ-45 connector.

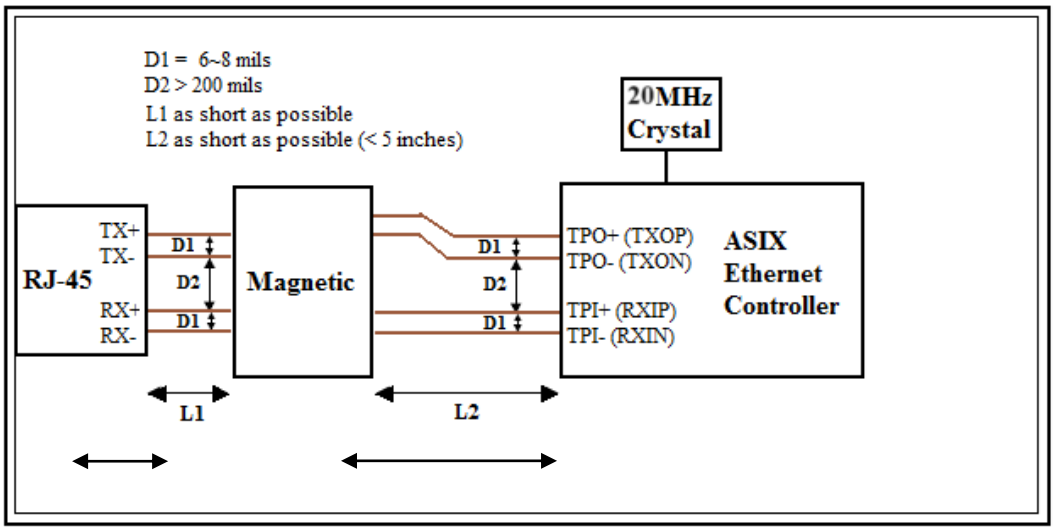
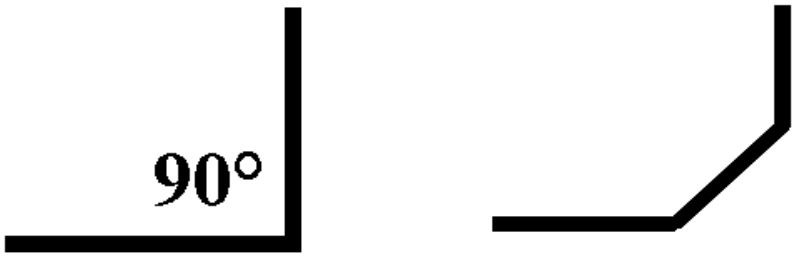


Figure 6. Ethernet TX± and RX± Differential Pairs Layout

1. The ideal Ethernet differential impedance between TX + and TX- traces (between RX+ and RX-) should be 100 Ohm. The Ethernet differential impedance between the TX+/TX-/RX+/RX- trace and GND should be 50 Ohm respectively. The designer can use the trace width of TX+/TX-/RX+/RX- traces to fine tune the Ethernet differential impedance value (a wider trace has a smaller impedance value). If the PCB manufacturer provides TDR measurement, it helps control the PCB Impedance.
2. The crystal/oscillator clock source and the switching noises from digital signals should be kept away from the TX± and RX± differential pairs.
3. The Ethernet magnetic should be placed as close to the RJ-45 connector as possible.
4. The Ethernet controller should be placed as close as possible to the magnetic. If there are some limitations on the PCB layout, the trace length from the Ethernet controller to the magnetic should not be longer than 5 inches.
5. The TX± and RX± differential pairs should be routed as close as possible. The trace spacing D1 between TX+ and TX- (or between RX+ and RX-) pair should be in 6 ~ 8 mils. The trace width should be adjusted accordingly to yield the required trace impedance.
6. The spacing D2 between the TX± and RX± differential pairs should be larger than 200 mils. If the PCB layout is really difficult to meet this requirement, the D2 spacing should be as larger as possible.
7. Route the TX± and RX± differential pairs as straight as possible and keep them in parallel for differential pairs.
8. Keep the trace length difference between the TX+ and TX- (or RX+ and RX-) pair within 700 mils.
9. Route the TX± and RX± differential pairs running symmetric, equal length and close whenever possible.

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10. Avoid using vias on the traces of the TX± and RX± differential pairs. If the PCB layout really needs to use vias on the differential pairs, please match the vias to keep the differential pairs balanced.
11. The power plane and digital ground plane should not be placed under the magnetic and RJ-45 connector.
12. Avoid routing the signal trace with right angle, instead, the signal trace should be routed with multiple 135° angles.



Worst Better

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11.Thermal Considerations

This section describes some information about how to reduce the operating temperature on the embedded Ethernet applications.

11-1 Improve the Cooling Plane

There are two major heat sources on the USB-to-LAN Ethernet applications. One is the Ethernet controller and the other is the external voltage regulator.

You can connect the VCC/GND pins of the USB-to-LAN Ethernet controller with wide traces to the respective power/ground planes to increase the cooling effect and reduce the operating temperature of the Ethernet controller.

You can also add a cooling plane for the external voltage regulator to reduce the operating temperature of the external voltage regulator.

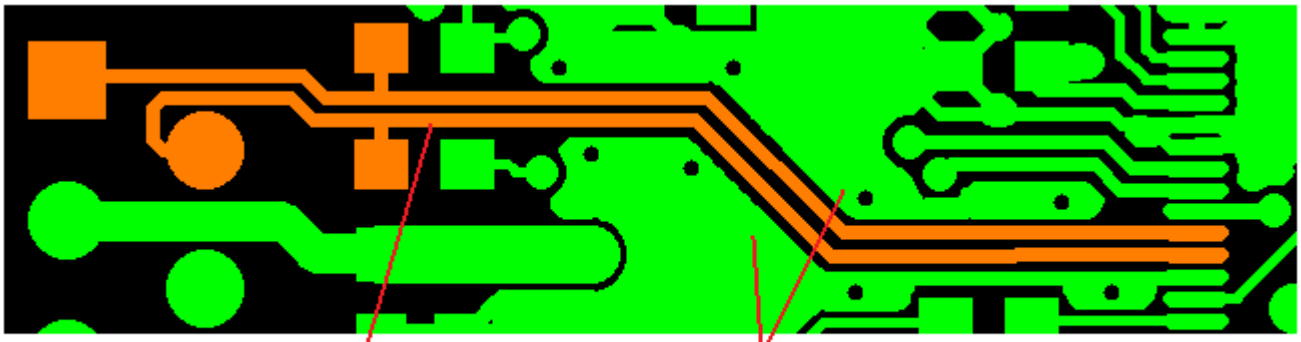
11-2 Improve the Air Convection

If the USB-to-LAN Ethernet controller is implemented on the embedded system, you can place the USB-to-LAN Ethernet controller at the location with good air convection and stay away from the high-operating-temperature IC such as voltage regulator or MCU to reduce the operating temperature of the Ethernet controller.

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12.EMI Considerations

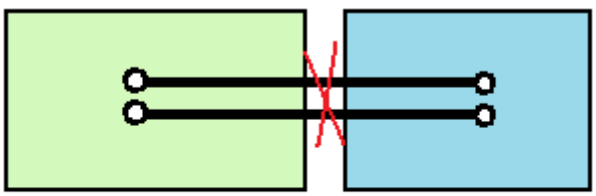
1. The high frequency signals traces such as USB/Ethernet differential signals, clock signals, etc. shall be surrounded by GND to prevent interference to other signal wires. Better EMI effect can be attained by a one-piece GND underneath the IC furnished with PTH holes to enlarge the GND area.



2. The chosen connector must be shielded so that EMI integrity of the design is not compromised. The shield must be electrically connected to chassis ground to extend the chassis barrier for high frequency emissions. If an unshielded connector were used, the EMI would pass through the nylon material of the connector. The shield will also prevent less external EMI from entering the chassis.

3. To reduce electromagnetic emissions and susceptibility, it is imperative that traces from the transceiver to the magnetic sand from the magnetic to the RJ-45 be routed as differential pairs. The objective is to close the loop area formed by the two conductors. The radiated field from the loop or the voltage picked up by the loop by external fields is governed by the field strength and the area formed by the two conductors. Reasonable board design uses 5~10 mils trace widths separated by 10 mils. Transmit differential pairs should be routed adjacent to a VDDO power plane.

4. DON'T CROSS ANY SIGNAL TRACES OVER ANY REFERENCE PLANE CUTS. This might cause some unpredictable EMI problems.



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5. It is essential to maintain the continuous stable ground plane when board design to protect EMI emissions. The PCB designer should look upon the path currents, which take on the ground reference plane, and reduce the loop area. The continuous ground plane helps provide for a low inductance signal return path, and helps maintain a stable power supply. Simply, the most effective method of reducing EMI causes is to decrease the loop area. The main target of ground plane is designed to carry DC current only. Do not use this area as a return path for high speed signals.

6. Designer can consider implementing the choke components on the Fast Ethernet differential signals traces to avoid the possible EMI issues if necessary.

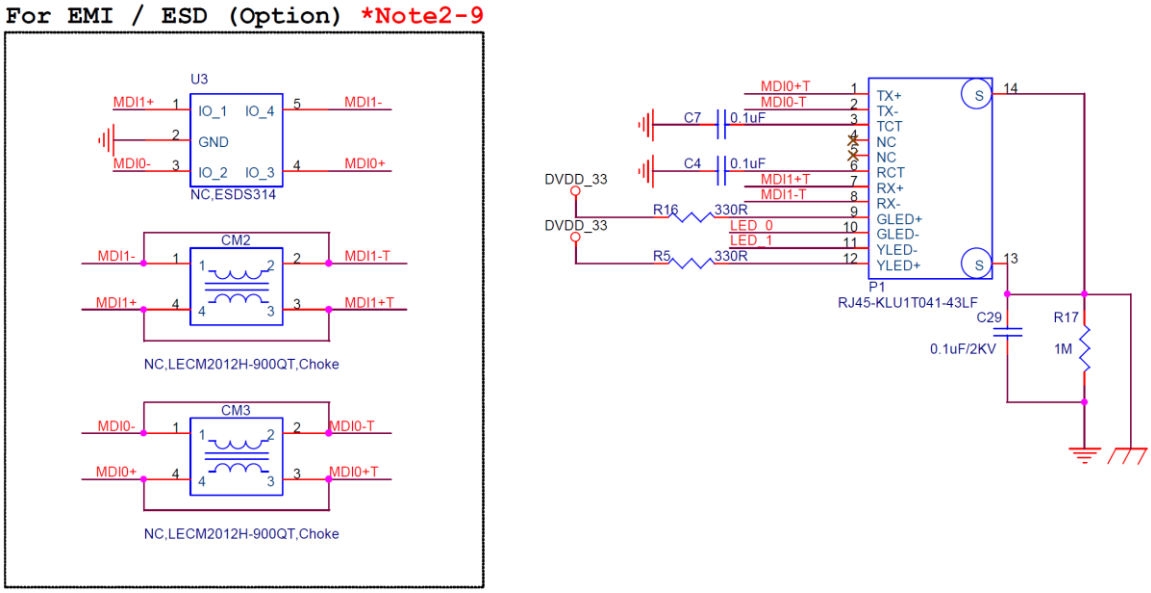


Figure 7. EMI Consideration Reference Circuit (Optional)

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13.ESD Considerations

This section describes some information about the ESD design guideline. Users can refer to the following circuit to avoid the ESD issue.

The high-energy pulses can cause abnormal system behavior and/or damage the silicon chips. The high energy can enter the system through the RJ-45 cable. There are eight wires in the RJ-45 cable and are grouped into 4 pairs as 2 active pairs (pin 1,2 & 3,6) and 2 unused pairs (pin 4,5 & 7,8).

If the high energy enters the system through the active pairs (pin 1,2 & 3,6) of RJ-45 cable, the Ethernet transformer will isolate the high energy. The unwanted high energy will be redirected to Chassis ground through a high voltage capacitor (2KV). If the high energy enters the system through the unused pairs (pin 4,5 & 7,8) of RJ-45 cable, these pins are connected to Chassis ground through a high voltage capacitor (2KV) so the unwanted high energy will be redirected to Chassis ground too.

Note: The following figures are exemplified magnetic circuit for the ESD considerations. Please refer to the respective AX88772E reference schematics for correct magnetic circuit.

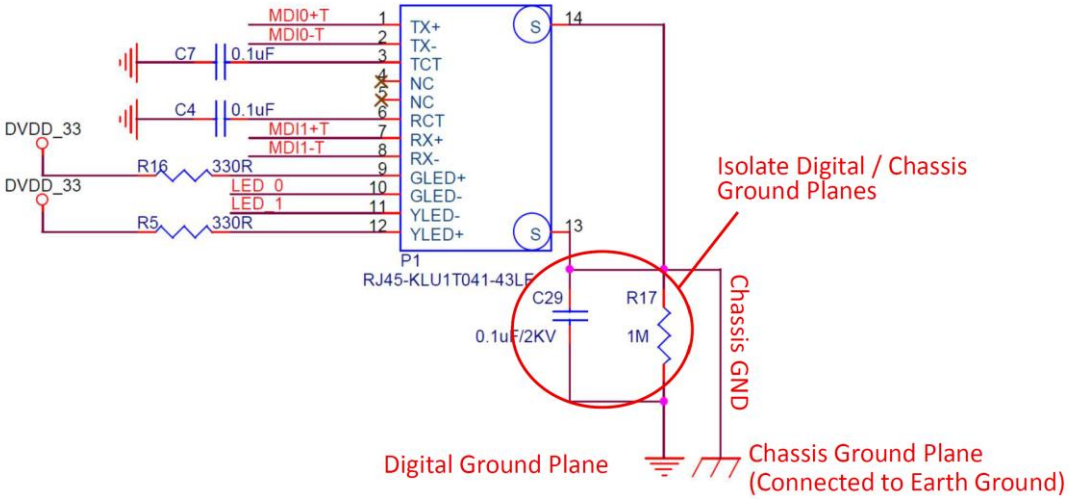


Figure 8. A Sample Single RJ-45 Connector Magnetic Circuit for ESD Considerations

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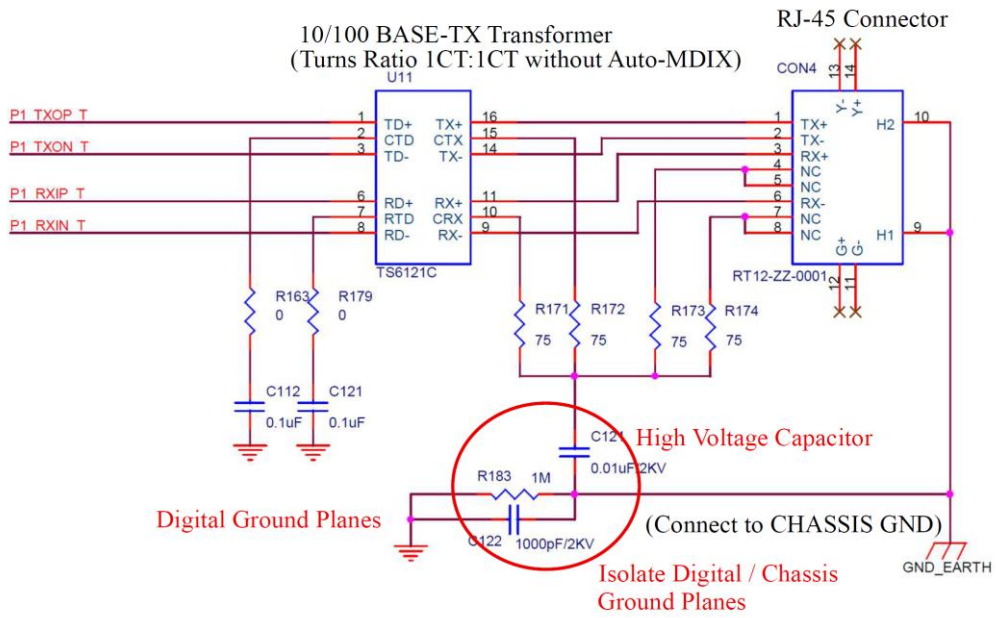


Figure 9. A Sample Separated Magnetic and RJ-45 Connector Circuit for ESD Considerations

You should isolate the chassis ground plane and digital ground plane through the 1M resistor and 0.1uF capacitor.

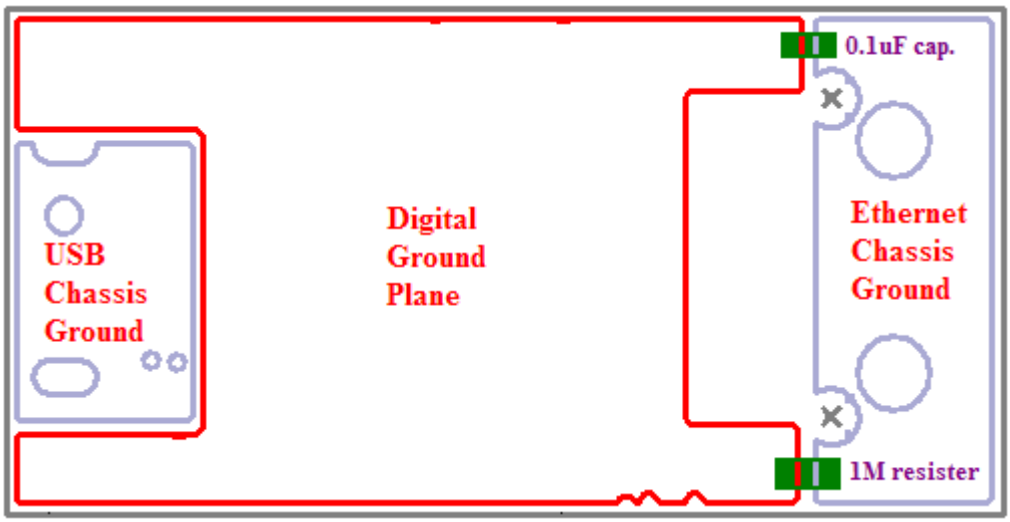


Figure 10. A Sample PCB Layout for ESD Considerations

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The fatal discharge may generate from a charged Ethernet cable, or out from outside devices such as PoE, or a human body, etc. Ethernet ICs become vulnerable to damage from Electrostatic Discharge (ESD) within those unpredictable factors. Protecting Ethernet interfaces from cable discharges can create a challenge for board designers. Be sure a good protective circuit must effectively fasten a transient to a safe voltage, and must present an acceptable capacitive load on high-speed differential transmission lines as well.

Below is a sample circuit for ESD protection. Please refer to the ESD Protection Diode datasheet for detailed ESD protection circuit.

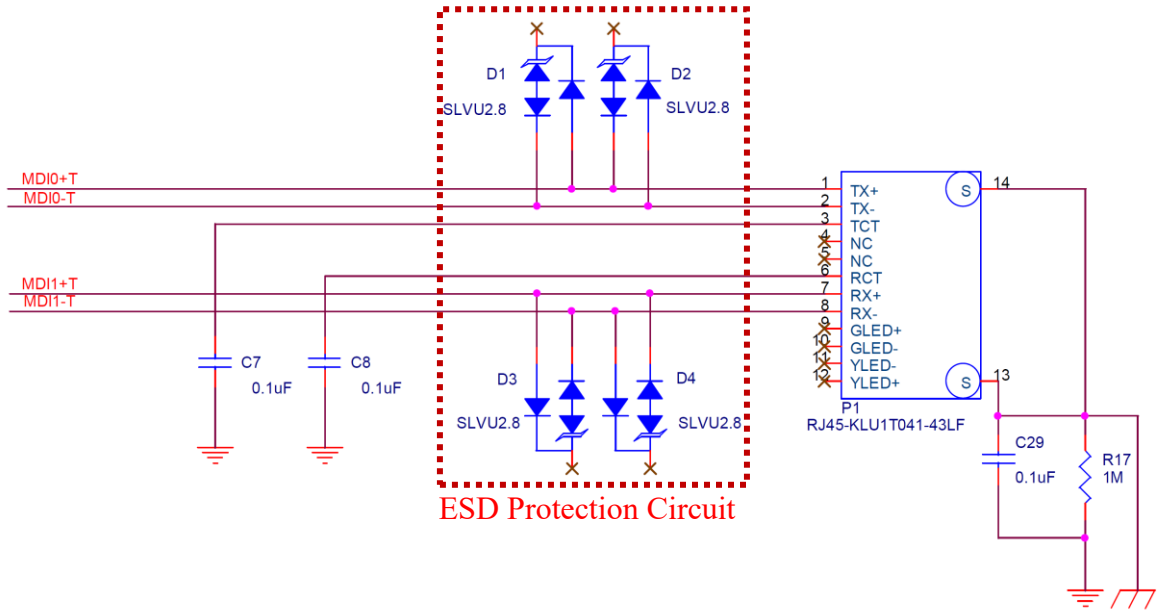


Figure 11. A Sample ESD Protection Circuit for Single RJ-45 with Integrated Magnetic (Optional)

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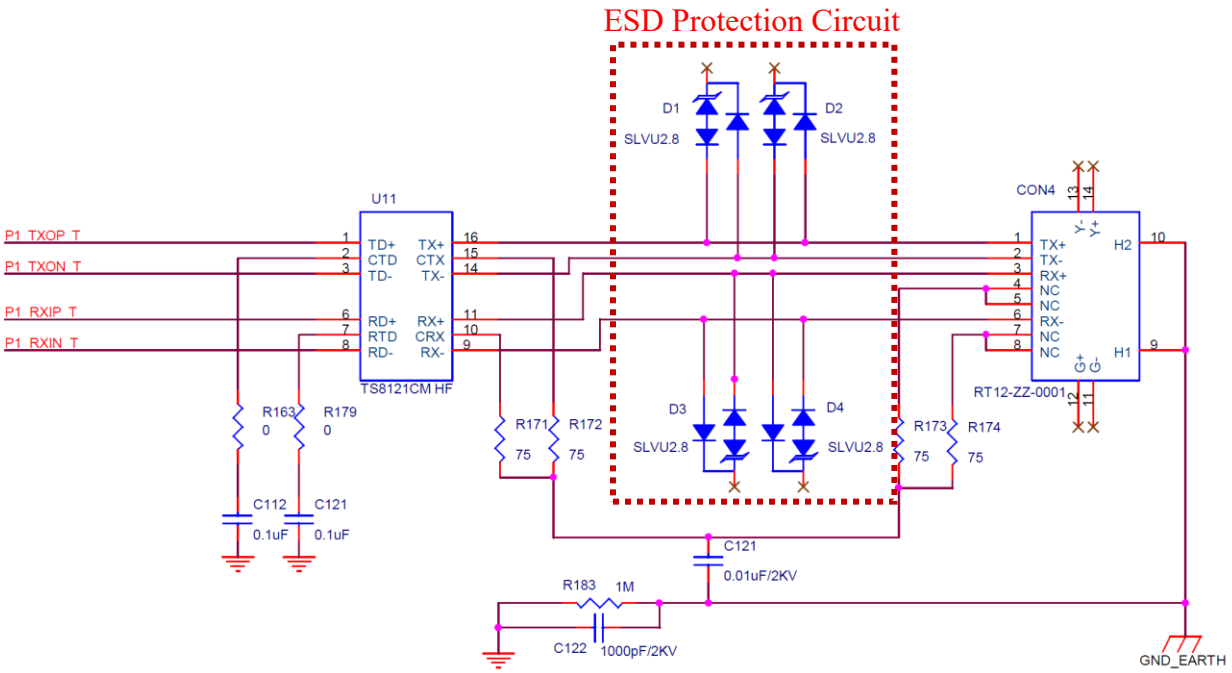


Figure 12. A Sample ESD Protection Circuit for Separate RJ-45 + Magnetic (Optional)

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